

BIOLOGY

Time: 3 hours

Marks: 70

Note:

- (a) All questions are compulsory.
- (b) Draw a neat labeled diagram wherever necessary.
- (c) Question paper consists of 3 questions. Divided into **FOUR** sections .A,B,C,D
- (d) SECTION –A contains 1to 4 questions of multiple choice of questions carrying 1 mark each and question 5 to 8 are very short answer type of questions carrying one mark each.
- (e) SECTION B contains 9 to 18 of short answer type of questions carrying two marks each. Internal choice is provided to only one question.
- (f) SECTION -C contains 19 to 27 of short answer type of questions carrying three marks each. Internal choice is provided to only one question.
- (g) SECTION -D contains 28 to 30 of long answer type of questions carrying five marks each. Internal choice is provided to each question.

SECTION –A

Q.1 As the base sequence on one strand of DNA decides the base sequence on other strand, this strand is considered as -----. (1)

- a) Descending Strand
- b) Leading strand
- c) Lagging strand
- d) Complementary Strand

Answer- complementary strand

Q.2 ----- Shows haplo- Diploid type of sex determination. (1)

- a) Pigeon
- b) Honey Bee
- c) Parrot
- d) Snake

Answer: (b)

Q.3 Membrane bound receptors and hormones produce second messenger like ----- (1)

- a) Renin
- b) IP_3
- c) ANF
- d) GFRC

Answer- IP_3

Q.4) during double fertilization second male gamete fuse with ----- (1)

- a) Antipodal cell
- b) Egg cell
- c) Secondary Nucleus
- d) Synergids.

Answer: Secondary Nucleus

Q.5. what is Sinus arrhythmias? (1)

Answer: During inspiration heart rate increases and during expiration it falls. This causes variation from normal rhythm of heart known as sinus arrhythmias.

Q.6 By which process ammonia is converted into urea? (1)

Answer: Conversion of ammonia into urea takes place by the process of ureotelism.

Q.7 Give the role of Plasmids in bacterial cell. (1)

Answer: Plasmid are the small extra chromosomal, double stranded circular form of DNA that replicate automatically.

Q.8 A person is showing symptoms like increased BMR, heart rate, pulse rate, blood pressure and deposition of fat in the eye socket. Name the disease he is suffering from. (1)

Answer: he is suffering from Grave's disease. It is caused by hyper secretion of Thyroxin hormone.

SECTION- B

Q.9 Define Apiculture. Name the product obtained from it. (2)

Answer: Apiculture deals with artificial rearing of honey bees to obtain bee's product like pollen, royal jelly, honey, wax. Honey is a food of high nutritive value. Bees are also help in pollination of various crops.

Q.10 Define Bio fertilizers. Give two types of fungal fertilizers. (2)

Answer: The fertilizers that use bio-agents like culture of blue green algae or other microorganism instead of chemicals rhizobium, azotobacter are used as biofertilizers. In order to make soil fertile and replenish its nutrient is known as biofertilizers.

Q.11 Give the types of blood protein and Human hormones produced by recombinant DNA. (2)

Answer: Blood protein produced by recombinant DNA is Erythropoietin, Factors vii, viii, ix, Tissue plasminogen activation, Urokinase.

Human hormone are epidermal growth factor, Follicle stimulating hormone, Insulin, Nerve growth factor, relaxin, and somatotropin.

Q. 12. Write two scientific and commercial values of transgenic animals in favour of Human beings. (2)

Answer: Most of the transgenic animals are rat. Scientific uses of transgenic animals are:

Transgenic animals are used to study how genes are regulated and how they affect the normal human body functions and its development.

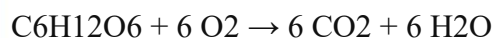
Commercial use of transgenic animals is following-

Transgenic animals are used to produce useful biological products. For example Rosie is the first transgenic Cow which produces protein enriched milk

Q.13. Define 'Respiratory Quotient'. Calculate the Respiratory Quotient for Carbohydrate (2)

Answer: The ratio of volume of CO₂ evolved to the volume of O₂ consumed in respiration is called the respiratory quotient.

Glucose metabolism breaks down in a 1:1 equivalent of oxygen to carbon dioxide. The equation shows that 6 oxygen molecules are used to produce 6 carbon dioxide molecule.



So respiratory Quotient for carbohydrate is 1.0.

Q.14. Light and dark reactions are interdependent. Explain (2)

During light reaction two compounds are formed when solar energy is converted into chemical energy. These are NADPH and ATP. In the dark reaction utilizing the energy from ATP and hydrogen from NADPH water combines with carbon dioxide to form carbohydrates. Thus solar energy is converted into chemical energy to form glucose during the light and dark reaction.

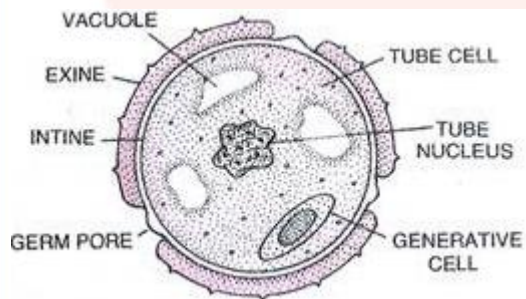
Q.15 Classify the chromosome on the basis of position of centromere. (2)

Answer: On the Basis of position of centromere chromosomes are divided into following types:

- 1) Metacentric: When the centromere is present in the middle of the chromosome the chromosome is called Metacentric.
- 2) Sub Metacentric: When the centromere is present slightly away from the middle. It is called the sub-metacentric chromosome.
- 3) Acrocentric: when centromere is present towards the end of the chromosome it is called acrocentric chromosome.
- 4) Telocentric: When the centromere is strictly confined to the end of the chromosome it is called telocentric.
- 5) Acentric: Centromere is absent in this type of chromosome.

Q.16. Sketch and Label structure of Male gametophyte in Angiosperm. Define Siphonogamy. (2)

Answer:



STRUCTURE OF MALE GAMETOPHYTE

Siphonogamy is a fertilization process in which non motile male gametes are transported up to the female gamete through a pollen tube.

Q.17. Match the following and rewrite.

(2)

Group ' A '

Group ' B '

a) Diethyl Carbamacine

1) AIDS

b) Widal test

2) Pneumonia

c) Albendazole

3) Filariasis

d) HAART

4) Typhoid

5) Ascariasis

Answer:

Group A

Group B

a) Diethyl Carbamacine

1) Filariasis

b) Widal test

2) Typhoid

c) Albendazole

3) Ascariasis

d) HAART

4) AIDS

Q.18. Complete the following Chart and rewrite.

(2)

S.No	Agencies	Type of pollination
1	Water	-----
2	-----	Entomophily
3	Bat	-----
4	-----	Ornithology

Answer-

S.No	Agencies	Type of pollination
1.	Water	Hydrophily
2.	Insect	Entomophily
3.	Bat	Chiropterophily
4.	Bird	Ornithology

SECTION-C

Q.19. What is Bio fortification? Explain selective breeding with suitable example.(3)

Answer: Bio fortification is a method of breeding crops to increase their nutritional value. It focuses on making plant food more nutritious.

For developing a new variety of crop plant by selective breeding, plants breeders search seeds for germ plasma banks for existing varieties of crops. Crops must be bred with sufficient amounts of nutrients to have a measurable positive impact on human health.

Q.20. In the light of Griffith's experiment, explain the action of two strains of Diplococcus Pneumonia and give his conclusion. [3]

Answer: Frederick Griffith observed two strains of D. Pneumoniae. The colony produced by one strain forms a capsule around the cell and has glistening appearance and is called smooth (S) type. The colony produced by other strain is encapsulated and has an irregular appearance and is called rough (R) type. The S-strain is virulent and can cause pneumonia as the capsule protect the bacteria from mammalian immunoglobulin defenses. The R-strain is non-virulent. When Griffith injected laboratory mice with a mixture of living a virulent R-type and heat killed S-type virulent, unexpected symptoms of pneumonia appeared, causing high mortality.

Q.21. Give scientific reason.

A) The pyramid of energy is always upright.

Answer- When the energy moves from one trophic level to the next level, there is a loss of some energy in the form of heat. Therefore the energy is always more on the lower trophic levels as compared to the higher trophic levels.

B) In an ecosystem energy flow is always unidirectional.

Answer: The energy captured by autotrophs does not go back to the solar input or the energy which passes to herbivore does not go back to the autotrophs so it is said that in an ecosystem energy flow is unidirectional.

C) Ozone present in stratosphere is called good ozone.

Answer: Ozone is present and formed in stratosphere. About 90% of ozone is present in stratosphere. Ozone present in Stratosphere is called good ozone as it protects all living things from harmful UV radiations. UV rays are highly injurious as it causes mutations.

Q.22 .Define reproductive isolation and explain two types of reproductive isolation. (3)

Reproductive isolation is brought about by the change in genetic material, gene pool and structure of genital organs. It prevents interbreeding between populations whether they are geographically isolated or living close together. Types of isolation are:

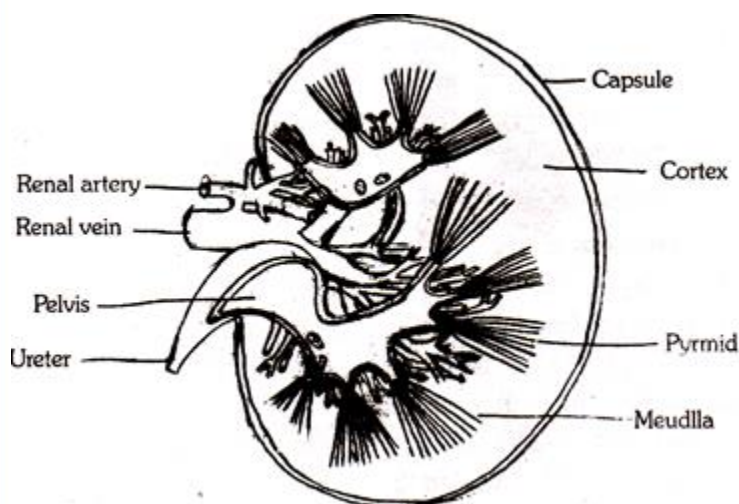
a) Seasonal isolation: Members of a population attains sexual maturity at the different times thus preventing interbreeding.

b) Mechanical Isolation: The members of two populations have differences in reproductive structures so interbreeding is not possible.

Q.23. Name the connecting link between Glycolysis and TCA cycle and explain it. (3)

Answer- Glycolysis and Kreb cycle are connected by link reaction or acetylation of pyruvate. Pyruvic acid undergoes oxidative decarboxylation before entering Kreb's Cycle resulting in the removal of Carbon dioxide with oxidation by removal of $2H^+$. Pyruvic acid undergoes decarboxylation first and then oxidation by removal of $2H^+$ and $NADH_2$ is formed. Acetyl Co-A is also formed. Glycolysis terminates with the formation of Pyruvate and enters as acetyl CoA in the Kreb's Cycle.

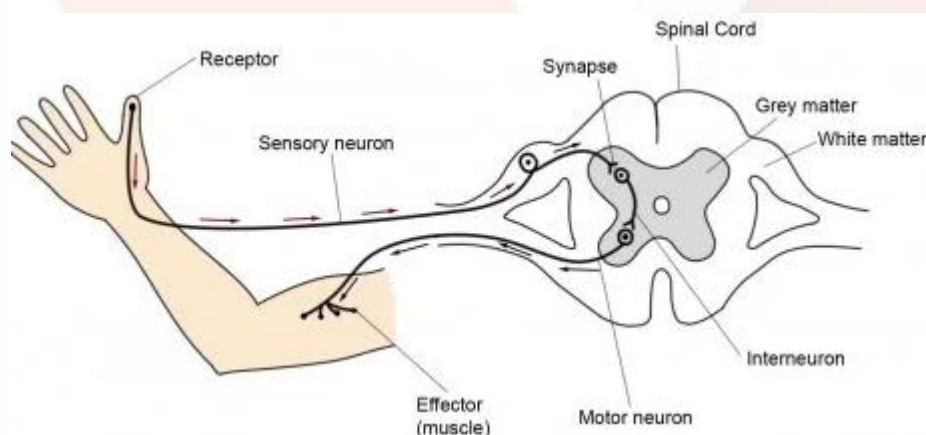
Q.24. Explain internal structure of Kidney with the help of suitable diagram.(3)



INTERNAL STRUCTURE OF KIDNEY

Kidney is the excretory organ and is situated below the diaphragm on the left and right sides. The right kidney is lower than the left kidney because the liver takes up a lot of space on that side. Each kidney consists of an outer cortex and inner medulla. They both contain uriniferous tubules or nephrons which are the structural and functional units of the kidney. The cortex is subdivided into alternating radial tracts known as rays and labyrinths. The renal medulla consists of the Loop of Henle, collecting tubules, and duct of Bellini.

Q.25 Explains the mechanism of reflex Action with the help of suitable diagram.(3)



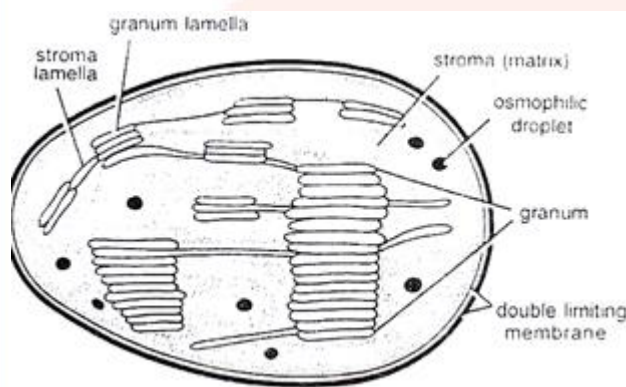
Answer: A quick, **Automatic**, involuntary and often unconscious action brought about when the receptors are stimulated by external or internal stimuli. Reflex action is controlled by the Central Nervous System. It is a structural and functional unit of reflex action.

Q.26. Define Pollution.” Industries are pouring poison in water.” -Explain (3)

Answer: Most of the river and freshwater streams are polluted by industrial wastes and effluents. Such pollutants come along waste waters of different industries such as petrochemical complexes, fertilizers factories, oil refineries, textile, sugar, rubber, plastic etc. Very few industries have effluent treatment facilities

Q.27. With the help of suitable diagram describe ultra-structure of cell organelle, which is essential for photosynthesis.

(3)



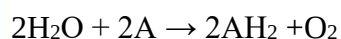
STRUCTURE OF CHLOROPLAST

Photosynthesis takes place in chloroplast. In higher plants lens shaped chloroplast are present. It is a double membrane structure and an outer and inner membrane collectively called as peristromium. It encloses colourless matrix called stroma. Embedded in the stroma are 40 to 60 green coloured structures called grana. Each granum consists of small disc like lamellae or thylakoid. Photosynthetic pigments are embedded in the thylakoids. Grana are interconnected by membrane called stroma lamellae which helps in rapid transport of materials.

OR

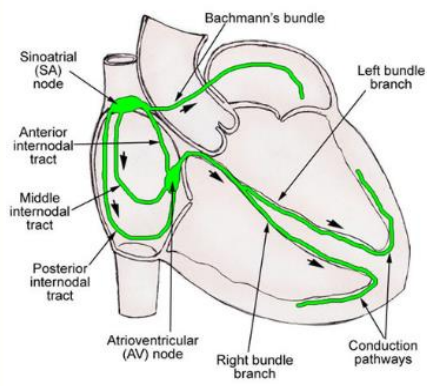
Q.27. During the process of Photosynthesis “O₂ is evolved from water and not from CO₂”. Give the experimental proof given by Robert Hill.

Robert hill performed an experiment. In which he suspended isolated chloroplast from spinach leaves in water which was without CO₂. He then added ferric salt as hydrogen acceptor and suspension was exposed in sunlight. He observed that ferric salts are reduced to ferrous and O₂ bubbles is evolved. Thus he proved that H₂O splits. It is shown as follows



SECTION-D

Q.28) Explain with the help of suitable diagram conducting system of Human Heart. (3)



CONDUCTING SYSTEM OF HUMAN HEART

Answer: Human Heart is myogenic. The heart beat originates in the modified cardiac muscles called Sino atrial node (SA node). The wave of contraction or cardiac impulse generated by SA node is conducted by cardiac muscle fibres to both the atria causes their contraction is located in the wall of right atrium called atria systole. The atrio ventricular node called AV node is located in the wall of right atrium and receives the wave of contraction generated by SA node through internodal pathways. Bundles of HIS arise from AV node and give rise to Purkinje fibres which penetrate into myocardium of ventricle causing their contraction.

OR

Q.28. Give reasons:

A) Lymphatic vessels are milky in appearance.

Answer: Lymphatic vessels when they pass through the intestine absorb fat and are called lacteals so they appear milky in appearance.

B) Monocytes are called scavengers.

Answer- Monocytes enlarge and differentiate into macrophages which engulf microorganisms and remove cell debris at the site of infection. Hence they are called scavengers.

c) The wall of left ventricle is thicker than right ventricle.

Answer- Left ventricle has a thicker wall than the right ventricle because it has to pump blood to all parts of the body.

(d) Valves are present in the veins.

Answer-Valves are present in the veins to prevent backward flow of blood.

(e) Pulmonary veins carry oxygenated blood.

Answer-The oxygenated blood from the lungs is brought to left atrium by pulmonary vein so it carries oxygenated blood.

Q.29) which phenomenon gives 2: 1 ratio instead of 3: 1 ratio? Describe with graphical Representation. (5)

Pleiotropic genes is a gene which controls two or more traits or expressions respectively and the phenomenon is known as pleiotrophy. For example the disease sickle cell anemia is caused by this phenomenon. This disease is caused by gene Hb^s . In a normal or healthy individual the gene is Hb^a which is dominant. A heterozygous is carrier and shows mild sign of anemia. The homozygous recessive individual shows fatal anemia and die. Hence if two carriers of sickle cell anemia mate, the children are formed in the ratio of 1: 2: 1 with normal 1 and 2 carrier and 1 diseased. Because the diseased child die the observed ratio is of the surviving children is 1: 2.

OR

Q.29) A pea plant homozygous for yellow round seed is crossed with its recessive parent. Calculate the phenotype and genotype ratio with the help of checker ratio.

Pea plant with homozygous yellow and round seed is represented by YYRR for and yyrr for homozygous green wrinkled seed pea plant

Y is represented for yellow flower and y is represented for green flower.

R is represented for round flower and r is represented for wrinkled seed.

Across between two homozygous parent gives F1 generation phenotype is Yellow round

	Yellow Round	Green wrinkle
Genotype	YYRR	yyrr
Gametes	YR	y
F1 generation phenotype	Yellow round	
(YyRr)		
Selfing of F1 Generation	YyRr	YyRr
Gametes	YR Yr yR yr	

	YR	Yr	yR	yr
YR	YYRR Yellow Round	YYRr Yellow Round	YyRR Yellow Round	YyRr Yellow Round
Yr	YYRr Yellow Round	YYrr Yellow Wrinkled	YyRr Yellow Round	Yyrr Yellow Wrinkled
yR	YyRR Yellow Round	YyRr Yellow Round	yyRR Yellow Round	yyRr Green Round
Yr	YyRr Yellow Round	Yyrr Yellow Wrinkled	yyRr Green Round	yyrr Green Wrinkled

The phenotypic ratio of F₂ generation is 9:3:3:1.

9- yellow round 3-yellow wrinkled 3- green round 1- green wrinkled

Q.30. After puberty human female shows cyclic changes in her reproductive system. Explain structural and Hormonal changes in the uterus. (5)

In human female reproductive system is immature at birth and develops at the puberty to be able to produce gametes, to carry out a foetus to full term. The uterus accommodate the embryo. Uterus produces vaginal and uterine secretion which helps the transport of sperm the fallopian tube for fertilization. Hormone released is oestrogen and progesterone. Oestrogen is released by Graffian follicle and corpus luteum secretes progesterone.

OR

Q.30. Give reasons:

(A) Scrotal sac serves as thermo regulator.

The scrotal sac act as a thermo regulator maintaining the testis at a temperature 2° to 2.5°C lower than that of body. This protects the sperm against high body temperature.

(B) Corpus Luteum gets converted into Corpus Albien in absence of fertilization.

Answer: Corpus Luteum is the temporary source of female hormone progesterone. If the ovum is not fertilized Corpus luteum degenerates and transform into a whitish scar called corpus albicans.

(C) Missing of menses is the first signal of pregnancy.

Answer: Menstruation occurs if there is no fertilization of ovum. The endometrium of uterus along with unfertilized egg is given out in the form of menstrual flow. The sloughing off uterine endometrium takes due to degeneration of Corpus Luteum. In the absence of functional Corpus Luteum progesterone level falls down.

(D) Surgical sterilization is the permanent method of birth control.

Answer: Surgical sterilization is a terminal method to prevent anymore pregnancy. Surgical interventions blocks gamete transfer and thereby prevent conception.

(E) Human egg is microlethical.

Answer: Microlethical eggs have much less amount of yolk than the amount of cytoplasm. These eggs are very small in size. Human eggs contain very little yolk so they are called microlethical.